



Critical Kernel Imperfectness in 4-quasi-transitive digraphs and 4-anti-transitive digraphs of small diameter[☆]

Germán Benítez-Bobadilla^a, Hortensia Galeana-Sánchez^b,
César Hernández-Cruz^{a,*}

^a Facultad de Ciencias, Universidad Nacional Autónoma de México, Ciudad Universitaria, Av. Universidad 3000, Ciudad de México, 04510, CDMX, México

^b Instituto de Matemáticas, Universidad Nacional Autónoma de México, Ciudad Universitaria, Av. Universidad 3000, Ciudad de México, 04510, CDMX, México

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ABSTRACT

A kernel in a digraph is an independent and absorbent subset of its vertex set. A digraph is critical kernel imperfect if it does not have a kernel, but every proper induced subdigraph does. In this article, we characterize asymmetrical 4-quasi-transitive and 4-transitive digraphs, as well as 2-anti-transitive, and asymmetrical 4-anti-transitive digraphs with bounded diameter, which are critical kernel imperfect.

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1. Preliminaries

All digraphs considered are finite, with neither parallel arcs nor loops. For general concepts we refer the reader to [4,5]. The reader who is familiar with basic digraph definitions and nomenclature may skip this section altogether.

Let D be a digraph with vertex set $V(D)$ and arc set $A(D)$. If u and v are vertices of D such that $(u, v) \in A(D)$, we say that u and v *adjacent*, that u *dominates* v , that v *absorbs* u , that v is an *out-neighbour* of v , and that u is an *in-neighbour* of v . We use the notation $u \rightarrow_D v$ when u dominates v , and $u \not\rightarrow_D v$ when this does not happen. The *out-neighbourhood* (resp. *in-neighbourhood*) of a vertex v , denoted $N_D^+(v)$ (resp. $N_D^-(v)$), is the set of all its out-neighbours (resp. in-neighbours). The *out-degree* (resp. *in-degree*) of a vertex v , denoted $d_D^+(v)$ (resp. $d_D^-(v)$) is the number of its out-neighbours (resp. in-neighbours), this is, $d_D^+(v) = |N_D^+(v)|$ (resp. $d_D^-(v) = |N_D^-(v)|$). The subscript indicating which digraph we are referring to, in the notation defined in this paragraph (and in the rest of this section), may be omitted when this information is clear from the context.

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* Corresponding author.

E-mail addresses: german@ciencias.unam.mx (G. Benítez-Bobadilla), hgaleana@matem.unam.mx (H. Galeana-Sánchez), chc@ciencias.unam.mx (C. Hernández-Cruz).

A subset S of $V(D)$ is *absorbent* if for every vertex u not in S there is a vertex $v \in S$ such that $(u, v) \in A(D)$. A subset I of $V(D)$ is *independent* or *stable* if every pair of vertices in I is non-adjacent. A *kernel* is a subset K of $V(D)$ which is independent and absorbent. A digraph such that every proper induced subdigraph has a kernel is *kernel-perfect* if it has a kernel, and *critical kernel imperfect* if not.

A *directed walk* W in a digraph D is a sequence of vertices $P = (x_0, \dots, x_n)$ such that $x_i \rightarrow x_{i+1}$ for each $i \in \{0, \dots, n-1\}$; the length of W is n . A *directed path* is a directed walk where all vertices are different, and a *directed cycle* is a directed walk where all vertices are different, except the first one which is the same as the last one. Since we only consider directed walks, we call them walks (resp. paths and cycles). A k -*path* (resp. k -*cycle*) is a path (resp. cycle) of length k . For vertices u and v of D , a uv -*path* is a path having u as its initial vertex and v as its last vertex. The *distance* from u to v in D , denoted $d_D(u, v)$, is the length of the shortest uv -path. We say that D is *strongly connected* or *strong* if for every pair of vertices u and v , there is a uv -path in D .

If X and Y are disjoint subsets of $V(D)$, then $X \rightarrow_D Y$ means that (x, y) is an arc of D for every vertex x of X and every vertex y of Y . If $X \rightarrow_D Y$ and $Y \rightarrow_D X$, then we write $X \leftrightarrow_D Y$. Whenever $X = \{x\}$ or $Y = \{y\}$, we write $x \rightarrow_D Y$ or $X \rightarrow_D y$ instead of $\{x\} \rightarrow_D Y$ or $X \rightarrow_D \{y\}$, respectively. If $X \rightarrow_D Y$ and there is no arc from Y to X , then we write $X \mapsto_D Y$. When there is no ambiguity, we omit the subscript D .

An arc (u, v) of a digraph D is *symmetrical* if (v, u) is also an arc of D , and *asymmetrical* otherwise. If every arc of D is asymmetrical we say that D is an *asymmetrical digraph*. Analogously, if every arc of D is symmetrical we say that D is a *symmetrical digraph*. For a digraph D , its *underlying graph* $U(D)$ is the graph with the same vertex set as D , and such that two vertices are adjacent in $U(D)$ if and only if they are adjacent in D .

A digraph D is *semicomplete* if for every pair of distinct vertices u and v , we have $u \rightarrow_D v$ or $v \rightarrow_D u$. A *tournament* is an asymmetrical semicomplete digraph. Let k be an integer, with $k \geq 2$. A digraph D is k -*quasi-transitive* if for every pair of vertices u and v of D , the existence of a k -path from u to v in D implies that there is an arc between u and v . A *quasi-transitive digraph* is a 2-quasi-transitive digraph. A digraph D is k -*transitive* if for every pair of vertices u and v of D , the existence of a k -path from u to v in D implies $u \rightarrow_D v$. A *transitive digraph* is a 2-transitive digraph. Trivially, every k -transitive digraph is a k -quasi-transitive digraph. In particular, every transitive digraph is a quasi-transitive digraph. Moreover, every semicomplete digraph is a quasi-transitive digraph.

A digraph is *locally semicomplete* if the neighbourhood of any vertex induces a semicomplete digraph. Another two interesting families arise when we only ask for either the in-neighbours or the out-neighbours of each vertex to induce a semicomplete digraph. A digraph D is *locally in-semicomplete* (resp. *out-semicomplete*) if the in-neighbourhood (resp. out-neighbourhood) of every vertex of D induces a semicomplete digraph. A digraph is *arc-locally in-semicomplete* (*arc-locally out-semicomplete*) if for any pair of adjacent vertices x and y , every in-neighbour (out-neighbour) z of x is adjacent to every in-neighbour (out-neighbour) w of y , or $w = z$.

A digraph D is k -*anti-transitive* if the existence of a k -path from vertex u to vertex v implies that (u, v) is not an arc of D . Notice that 2-anti-transitive digraphs arise naturally as those digraphs not having a transitive tournament on 3 vertices as a subgraph.

For an integer m , with $m \geq 2$, and a nonempty subset J of $\mathbb{Z}_m - \{0\}$, the *circulant digraph* $\vec{C}_m(J)$ is defined to have vertex set $V(\vec{C}_m(J)) = \mathbb{Z}_m$ and arc set $A(\vec{C}_m(J)) = \{(i, j) : i, j \in \mathbb{Z}_m, j - i \in J\}$. There are two cases of circulant digraphs of special interest to us. Let n be an integer, with $n \geq 3$. The directed cycle $\vec{C}_n$ is the circulant digraph $\vec{C}_n(\{1\})$. Clearly, the complement of a circulant digraph is also a circulant digraph; complements of directed cycles are relevant for this subject. The *directed n -antihole* $\vec{A}_n$ is the circulant digraph $\vec{C}_n(J)$, where $J = \{1, \dots, n-2\}$; note that $\vec{A}_3 = \vec{C}_3$. A *directed antihole* is an directed n -antihole for some integer n . The family of directed antiholes is denoted by $\mathcal{A}$.

As is standard, for a nonempty subset S of $V(D)$, we denote by $D[S]$ the subdigraph of D induced by S . For digraphs D and H , we say that D is H -free if D does not contain H as an induced subdigraph. If $\mathcal{F}$ is a set of digraphs, we say that a digraph D is $\mathcal{F}$ -free if it is F -free for every $F \in \mathcal{F}$.

For digraphs D and D' , we use $D \cong D'$ to denote the existence of an isomorphism between D and D' .

2. Introduction

Kernels in digraphs were introduced by von Neumann and Morgenstern in the context of game theory [23], and received a lot of attention at the end of the twentieth century for their relation to the Strong Perfect Graph Conjecture (see e.g. [8]), now the Strong Perfect Graph Theorem [10]. Although kernels are useful to model a wide variety of problems, it is computationally hard to determine whether a digraph has a kernel. Chvátal proved that the recognition of digraphs that have a kernel is an NP-complete problem [11]. Furthermore, Hell and Hernández-Cruz proved that this problem remains NP-complete even when the underlying graph of the input digraph is 3-colourable [20] (this is in contrast to the fact that every bipartite digraph has a kernel). For this reason, it is interesting to find sufficient conditions, which can be efficiently verified, implying the existence of a kernel in a digraph.

As classical examples of critical kernel imperfect and kernel-perfect digraphs, we have that directed odd cycles are critical kernel imperfect digraphs, and directed even cycles are kernel-perfect digraphs. It is clear from the definitions that a digraph is kernel-perfect if and only if it does not contain a critical kernel imperfect digraph as an induced subdigraph. For this reason, for a family of digraphs $\mathcal{D}$, the knowledge of the exact digraphs in $\mathcal{D}$ which are critical kernel imperfect gives us a sufficient condition for a digraph in $\mathcal{D}$ to have a kernel. Moreover, if the number of critical kernel imperfect

digraphs in $\mathcal{D}$ is finite, a polynomial time algorithm for recognizing kernel perfect members of $\mathcal{D}$ trivially exists. Therefore, it is useful to exhibit the exact set of critical kernel imperfect digraphs in a family of digraphs. This has been done for some well-known families of digraphs, which we now survey.

Regarding small digraphs, in [2], Balbuena, Guevara, and Olsen characterized the asymmetrical critical kernel imperfect digraphs with at most 7 vertices as follows.

Theorem 1 ([2]). *There are exactly four asymmetrical critical kernel imperfect digraphs of order less than 8, namely, $\vec{C}_3$, $\vec{C}_5$, $\vec{C}_7$, and $\vec{C}_7(\{1, 2\})$.*

In [16], Galeana-Sánchez and Neumann-Lara characterized the semicomplete critical kernel imperfect digraphs as follows.

Theorem 2 ([16]). *A semicomplete digraph is critical kernel imperfect if and only if it is a directed antihole. Therefore, a semicomplete digraph is kernel-perfect if and only if it is $\vec{A}$ -free.*

A natural step after semicomplete digraphs, is to consider generalizations of tournaments (or more precisely, generalizations of semicomplete digraphs). One of the better known such generalizations is the family of locally-semicomplete digraphs. The critical kernel imperfect digraphs which are locally in-semicomplete, locally out-semicomplete or locally semicomplete digraphs, were characterized by Galeana-Sánchez and Olsen in [17].

Theorem 3 ([17]). *Let D be a digraph. If D is locally semicomplete, locally in-semicomplete, or locally out-semicomplete, then D is critical kernel imperfect if and only if it is a directed odd cycle, a directed antihole, or $D \cong \vec{C}_7(1, 2)$.*

Arc-locally semicomplete digraphs were introduced by Bang-Jensen in [3], as a common generalization of semicomplete and semicomplete bipartite digraphs. Critical kernel imperfect digraphs which are arc-locally semicomplete were characterized by Galeana-Sánchez in [14].

Theorem 4 ([14]). *Let D be a digraph. If D is arc-locally semicomplete, then it is critical kernel imperfect if and only if it is a directed odd cycle or a directed antihole.*

Wang characterized in [24] critical kernel imperfect digraphs which are arc-locally out-semicomplete and also critical kernel imperfect digraphs which are arc-locally in-semicomplete.

Theorem 5 ([24]). *Let D be an arc-locally (out-)in-semicomplete digraph. If D is a critical kernel imperfect digraph, then D is a directed odd cycle or a directed antihole.*

Quasi-transitive digraphs are also traditionally considered among the generalizations of semicomplete digraphs. The family of quasi-transitive digraphs was introduced by Ghouila-Houri in [18] to characterize comparability graphs as those graphs that admit a quasi-transitive orientation. In [6], Bang-Jensen and Huang gave a recursive characterization of quasi-transitive digraphs, which has led to solutions of many (usually difficult) problems. In [22], Hernández-Cruz and Galeana-Sánchez introduced k -transitive and k -quasi-transitive digraphs, generalizing the notions of transitive and quasi-transitive digraphs, respectively. The properties of quasi-transitive digraphs and their generalizations have been studied by several authors. In [15], a thorough compendium dedicated to quasi-transitive digraphs and their extensions can be found.

In [17], Galeana-Sánchez and Olsen analysed those digraphs having an underlying perfect graph. Since underlying graphs of quasi-transitive digraphs are comparability graphs, the following result completely characterizes the critical kernel imperfect quasi-transitive digraphs.

Theorem 6 ([17]). *Let D be a digraph. If the underlying graph of D is a perfect graph, then D is critical kernel imperfect if and only if D is a directed antihole.*

Also in [17], Galeana-Sánchez and Olsen studied asymmetric critical kernel imperfect 3-quasi-transitive digraphs. The analysis for the general case (not necessarily asymmetric) was finished by Wang in [24].

Theorem 7 ([24]). *Let D be a 3-quasi-transitive digraph. If D is a critical kernel imperfect digraph, then D is a directed antihole.*

Recently, k -anti-transitive digraphs have gained popularity in the context of Seymour's Second Neighbourhood Conjecture. Seymour's Second Neighbourhood Conjecture has been proved for 2-anti-transitive digraphs [9], 3- and 4-anti-transitive digraphs [13], 5-anti-transitive oriented graphs [12], and 6-anti-transitive oriented graphs [19]. It is surprising that these nice results have been proved, even in the absence of a description of the global structure of k -anti-transitive digraphs. In the present work, we characterize critical kernel imperfect digraphs for some subclasses of 2- and 4-anti-transitive digraphs, providing further evidence that these families may have a rich structure which is worth studying.

In this work, we characterize critical kernel imperfect digraphs in some well-established families of digraphs. In the context of k -quasi-transitive digraphs, we prove that there are exactly two asymmetrical 4-quasi-transitive digraphs, namely $\vec{A}_3$ and $\vec{C}_5$. Afterwards, the general case for 4-transitive digraphs is considered, and we show that the only 4-transitive critical kernel imperfect digraphs are $\vec{A}_3$ and $\vec{A}_4$. Finally, we consider anti-transitive digraphs, showing that there are unique critical kernel imperfect digraphs for the families of the 2-anti-transitive digraphs of diameter 2, and the asymmetrical 4-anti-transitive digraphs of diameter 3.

The rest of the article is organized as follows. We finish this section by stating some basic known results on kernel-perfect and critical kernel imperfect digraphs, as well as the main results of the present work. In Section 3, a characterization of asymmetric critical kernel imperfect 4-quasi-transitive digraphs is presented. Critical kernel imperfect 4-transitive digraphs are characterized in Section 4. Antitransitive digraphs are considered in Section 5; critical kernel imperfect 2-anti-transitive and 4-anti-transitive of diameter 3 are characterized. Conclusions and open problems are presented in the final section.

It is clear from the definition that critical kernel imperfect digraphs are minimal digraphs without a kernel, this is, every digraph without a kernel contains a critical kernel imperfect digraph. From this observation, the following proposition, which is used extensively in the remainder of this work without explicit mention, is easily obtained.

Proposition 8. *If D is a critical kernel imperfect or a kernel-perfect digraph, then D does not properly contain a critical kernel imperfect subdigraph.*

We recall a result of Berge and Duchet from [7], which is also used many times in this work without explicit mention.

Theorem 9 ([7]). *If D is a critical kernel imperfect digraph, then D is strong.*

The main result of Section 3, dealing with asymmetrical 4-quasi-transitive digraphs, is the following.

Theorem 10. *The unique asymmetrical 4-quasi-transitive critical kernel imperfect digraphs are $\vec{A}_3$ and $\vec{C}_5$.*

Regarding 4-transitive digraphs, the main result of Section 4 is the following.

Theorem 11. *If D is a 4-transitive critical kernel imperfect digraph, then $D \cong \vec{A}_3$ or $D \cong \vec{A}_4$.*

Finally, in Section 5, we obtain the following results for 2-anti-transitive digraphs of diameter 2 and asymmetrical 4-anti-transitive digraphs of diameter 3.

Theorem 12. *The unique 2-anti-transitive critical kernel imperfect digraph of diameter 2 is $\vec{A}_3$.*

Theorem 13. *The unique asymmetrical 4-anti-transitive critical kernel imperfect digraph of diameter 3 is $\vec{C}_5$.*

3. 4-quasi-transitive digraphs

The following results describe the structure of k -quasi-transitive digraphs of diameter at least $k + 2$. In the first of the results, the case when k is even, is due to Wang and Zhang [25], and the case when k is odd is due to Alva-Samos and Hernández-Cruz [1].

Theorem 14 ([1,25]). *Let $k \geq 3$ be an integer and let D be a k -quasi-transitive digraph. Suppose that $P = (x_0, \dots, x_r)$ is a shortest path from x_0 to x_r in D , with $r \geq k + 2$.*

1. *If k is even, then $D[V(P)]$ is a semicomplete digraph and $x_j \rightarrow x_i$ for $1 \leq i + 1 < j \leq r$.*
2. *If k is odd, then $D[V(P)]$ is either a semicomplete digraph and $x_j \rightarrow x_i$ for $1 \leq i + 1 < j \leq r$, or $D[V(P)]$ is a semicomplete bipartite digraph and $x_j \rightarrow x_i$ for $1 \leq i + 1 < j \leq r$ and $i \not\equiv j \pmod{2}$.*

Lemma 15 ([25]). *Let k be an even integer with $k \geq 4$, and let D be a strong k -quasi-transitive digraph. Suppose that $P = (v_0, v_1, \dots, v_{k+2})$ is a shortest path from v_0 to v_{k+2} in D . For any $v \in V(D) - V(P)$, if $(v, V(P)) \neq \emptyset$ and $(V(P), v) \neq \emptyset$, then either v is adjacent to every vertex of $V(P)$, or $\{v_{k+2}, v_{k+1}, v_k, v_{k-1}\} \mapsto v \mapsto \{v_0, v_1, v_2, v_3\}$. In particular, if $k = 4$, then v is adjacent to every vertex of $V(P)$.*

Theorem 16 ([25]). *Let k be an even integer with $k \geq 4$, and let D be a strong k -quasi-transitive digraph. If $P = (v_0, \dots, v_{k+2})$ is a shortest path from v_0 to v_{k+2} , then the subdigraph induced by $V(D) - V(P)$ is a semicomplete digraph.*

It follows from Theorems 14 and 16 and Lemma 15 that a 4-quasi-transitive digraph of diameter at least 6 is a semicomplete digraph. Also, by Theorem 2, the unique semicomplete critical kernel imperfect digraphs are directed antiholes, which have diameter 2. Hence, the following result follows.

Corollary 17. *There are no critical kernel imperfect 4-quasi-transitive digraphs of diameter at least 6.*

In view of [Corollary 17](#), and since digraphs of diameter 1 are complete, and thus kernel-perfect, we focus on 4-quasi-transitive digraphs of diameter k , with $2 \leq k \leq 5$.

Lemma 18. *The unique asymmetrical critical kernel imperfect digraph of diameter 2 is $\vec{A}_3$.*

Proof. Let D be an asymmetrical critical kernel imperfect digraph of diameter 2, and let x be a vertex of D . Consider $S_1 = \{v \in V(D) : d(v, x) = 1\}$ and $S_2 = \{v \in V(D) : d(v, x) = 2\}$. Since D does not have a kernel, S_2 is nonempty. Also, since D is strong, it follows that $N^+(x) \neq \emptyset$. Moreover, the fact that D is asymmetrical implies $N^+(x) \subseteq S_2$. Thus, for $w \in N^+(x)$, there is $u \in S_1$ such that (w, u, x) is a path in D . Thus, (w, u, x, w) is a 3-cycle in D , and being D asymmetrical, we conclude $D[\{w, u, x, w\}] \cong \vec{A}_3$. It follows from the critical kernel imperfectness of D that $D \cong \vec{A}_3$. $\square$

Proposition 19. *There is no 4-quasi-transitive critical kernel imperfect digraph of diameter 2 such that every 3-cycle has at least 2 symmetrical arcs.*

Proof. Proceeding by contradiction, suppose that there is a critical kernel imperfect 4-quasi-transitive digraph D of diameter 2 such that every 3-cycle has at least 2 symmetrical arcs.

Let $v \in V(D)$ and $D_1 = D - v$. Since D is a critical kernel imperfect digraph, D_1 has a kernel N' . Observe that $N' \cup \{v\}$ is not an independent set, otherwise $N' \cup \{v\}$ would be a kernel of D . Moreover, if there is an arc from v to some vertex of N' , then N' is a kernel of D , which is impossible. Thus, we can assume that there is a vertex z of N' such that $(z, v) \in A(D)$ and there is no arc from v to some vertex in N' .

Let S_1 and S_2 be defined as $S_1 = \{z \in N' : d(z, v) = 1\}$ and $S_2 = N' - S_1$. By definition, $S_2 \cup \{v\}$ is an independent set. Consider $B = \{u \in V(D) : u \rightarrow_D z, u \rightarrow_{D_1} v \text{ and } u \rightarrow_{D_1} x, \text{ for some } z \in S_1 \text{ and for all } x \in S_2\}$. Note that $B \neq \emptyset$, otherwise $S_2 \cup \{v\}$ is a kernel of D . Moreover, let S_0 be the subset of S_1 such that $S_0 = \{z \in S_1 : \text{there is } u \in B \text{ such that } u \rightarrow_D z\}$, observe that $S_0 \neq \emptyset$.

Claim 1. *If $v \rightarrow_D w$ and there is $z \in S_1$ such that $w \rightarrow_D z$, then $v \leftrightarrow_D w$ and $w \leftrightarrow_D z$.*

Proof of Claim 1. Let $w \in N^+(v)$ and $z \in S_1$ such that $w \rightarrow_D z$. It follows that (v, w, z, v) is a 3-cycle of D . Even more, by hypothesis, and since $z \mapsto v$, then $v \leftrightarrow_D w$ and $w \leftrightarrow_D z$. $\blacksquare$

As a consequence of [Claim 1](#), and by definition of B , we have that if $u \in B$, then $\{v, u\}$ is an independent set.

Claim 2. $S_2 = \emptyset$.

Proof of Claim 2. Suppose, for the sake of contradiction, that there is $x \in S_2$. Since the diameter of D is 2, there is $y \in V(D)$ such that (v, y, x) is a path in D . As N' is a kernel and by definition of B , it follows that $y \notin S_1$ and $y \notin B$. Let $u \in B$. It follows that there is $z \in S_0$ such that $u \rightarrow_D z$. Moreover, (u, z, v, y, x) is a 4-path in D . Since D is 4-quasi-transitive, there is an arc between u and x . By definition of B , $(u, x) \notin A(D)$, thus $x \rightarrow_D u$. Now, since the diameter of D is 2, there is $w \in V(D)$ such that (v, w, z) is a path in D . By [Claim 1](#), we have that $v \leftrightarrow_D w$ and $w \leftrightarrow_D z$. Hence, (x, u, z, w, v) is a 4-path in D , and by hypothesis there is an arc between x and v , which is impossible by definition of S_2 . Therefore $S_2 = \emptyset$. $\blacksquare$

Claim 3. *If $x \in V(D) \setminus (\{v\} \cup S_0 \cup B)$, then $x \rightarrow_D v$.*

Proof of Claim 3. Let $x \in V(D) \setminus (\{v\} \cup S_0 \cup B)$. On the one hand, if $x \in N'$, then, by [Claim 2](#), we have $x \in S_1 \setminus S_0$. Thus $x \rightarrow_D v$. On the other hand, if $x \in V(D) \setminus N'$, then there is $y \in N'$ such that $x \rightarrow_D y$. By [Claim 2](#), we conclude $y \in S_1$. Since $x \notin B$, it follows that $x \rightarrow_D v$. $\blacksquare$

Claim 4. $|S_0| = 1$.

Proof of Claim 4. Observe that $S_0 \neq \emptyset$. Proceeding by contradiction suppose that $|S_0| \geq 2$. Let $z_1, z_2 \in S_0$. By definition of S_1 , $z_i \mapsto v$ with $i \in \{1, 2\}$. Since the diameter of D is 2, there is $w_i \in V(D)$ such that (v, w_i, z_i) is a path in D , for each $i \in \{1, 2\}$. In addition, by [Claim 1](#), we have that $v \leftrightarrow_D w_i$ and $w_i \leftrightarrow_D z_i$. If $w_1 \neq w_2$, then (z_1, w_1, v, w_2, z_2) is a 4-path in D . Since D is 4-quasi-transitive, there is an arc between z_1 and z_2 which is impossible because N' is an independent set. Hence, we can assume that $w_1 = w_2$. As $z \in S_0$, then there is $u \in B$ such that $u \rightarrow_D z_1$. It follows that (u, z_1, w, z_2, v) is a 4-path in D . Since D is a 4-quasi-transitive digraph, there is an arc between u and v , which is a contradiction. Therefore, $|S_0| = 1$. $\blacksquare$

Claim 5. $|B| = 1$.

Proof of Claim 5. Suppose, for the sake of contradiction, that there are distinct $u_1, u_2 \in B$. Since the diameter of D is 2, there is $y_i \in V(D)$ such that (v, y_i, u_i) is a path in D , for each $i \in \{1, 2\}$. By definitions of S_0 and B , we have $y_i \notin (S_0 \cup B)$. By Claim 3, we have that $y_i \leftrightarrow_D v$, for each $i \in \{1, 2\}$. It follows that (u_2, z, v, y_1, u_1) is a 4-path in D , where z is the unique vertex in S_0 . Since D is a 4-quasi-transitive digraph, there is an arc between u_1 and u_2 . Suppose without loss of generality that $u_2 \rightarrow_D u_1$. As the diameter of D is 2, there is $w \in V(D)$ such that (v, w, z) is a path in D , by Claim 1, $v \leftrightarrow_D w$ and $w \leftrightarrow_D z$. We have that (u_2, u_1, z, y, v) is a 4-path in D . Again, D is a 4-quasi-transitive digraph, so there is an arc between u_2 and v , contradicting the fact that $\{v, u_2\}$ is an independent set. Therefore, $|B| = 1$. ■

To finish the proof, let u be the unique vertex in B . We prove that $N = \{v, u\}$ is a kernel of D , reaching the contradiction we are looking for. By Claim 1, $\{v, u\}$ is an independent set. Let x be a vertex in $V(D) \setminus N$. If $S_0 = \{x\}$, then, by definition of S_0 , $x \rightarrow_D v$. If $x \in V(D) \setminus (\{v\} \cup S_0 \cup N)$, then, by Claim 3, we have that $x \rightarrow_D v$. Hence, N is an absorbent set, and in consequence, a kernel of D , contradicting the fact that D is critical kernel imperfect. Therefore, we conclude the desired result. □

Observe that every directed n -antihole, with $n \geq 4$, is a critical kernel imperfect digraph with a 3-cycle with exactly one symmetrical arc. Therefore, the hypothesis of Proposition 19 is sharp.

Lemma 20. Let D be an asymmetrical 4-quasi-transitive digraph of diameter 3. If D is strong and $\vec{A}_3$ -free, then D is locally semicomplete.

Proof. Let $v \in V(D)$. We prove that D is locally in-semicomplete and locally out-semicomplete. Since D is strong, we have $N^+(v) \neq \emptyset$ and $N^-(v) \neq \emptyset$.

Let $u \in N^+(v)$. By hypothesis, $1 \leq d(u, v) \leq 3$. Since D is an asymmetrical digraph, it is clear that $d(u, v) < 1$, moreover, $d(u, v) \neq 2$, otherwise D has an induced copy of $\vec{A}_3$ as a subdigraph. Hence, $d(u, v) = 3$ for all $u \in N^+(v)$. Let $u_1, u_2 \in N^+(v)$. From above, there is a path T , from u_1 to v of length 3. It follows that $T \cup (v, u_1)$ is a 4-path from u_1 to u_2 . Since D is 4-quasi-transitive, there is an arc between u_1 and u_2 . Hence $D[N^+(v)]$ is a semicomplete digraph. Moreover, since D is asymmetrical and has no $\vec{A}_3$ as induced subdigraph, $D[N^+(v)]$ is a transitive tournament. Therefore, D is a locally out-semicomplete digraph.

Now, let $w \in N^-(v)$. Following an argument analogous to the one in the previous paragraph, we conclude that $D[N^-(v)]$ is a transitive tournament. Therefore, D is a locally in-semicomplete digraph. Hence, D is a locally semicomplete digraph. □

Lemma 21. Let D be an asymmetrical 4-quasi-transitive digraph. If D has diameter 3, then it is not critical kernel imperfect.

Proof. If there is a 4-quasi-transitive critical kernel imperfect digraph D of diameter 3, then D is strong and has no $\vec{A}_3$ as induced subdigraph. Hence, by Lemma 20, D is a locally semicomplete digraph. From Theorem 3, it follows that D is either a directed odd cycle, a directed antihole, or $D \cong \vec{C}_7(1, 2)$. But $\vec{A}_3$ and $\vec{C}_5$ have diameter different from 3; directed odd cycles with at least 7 vertices and $\vec{C}_7(1, 2)$ are not 4-quasi-transitive digraphs, and $\vec{A}_n$ with $n \geq 4$ has symmetrical arcs. Therefore, there is no asymmetrical critical kernel imperfect 4-quasi-transitive digraph of diameter 3. □

The following lemmas describe some structural properties of asymmetrical $\{\vec{C}_3, \vec{C}_5\}$ -free 4-quasi-transitive digraphs of diameter 4 or 5.

Lemma 22. Let D be an asymmetrical 4-quasi-transitive digraph of diameter 4 or 5. Let v be a vertex of D and S_i be the set $\{x \in V(D) : d(x, v) = i\}$ with $i \in \{1, \dots, \text{diam}(D)\}$. If D is $\{\vec{C}_3, \vec{C}_5\}$ -free, then the following properties hold.

1. $S_1 = N^-(v)$.
2. For every $i \in \{2, \dots, \text{diam}(D)\}$, $S_i \rightarrow_{D^c} v$.
3. For every $i \in \{1, \dots, \text{diam}(D) - 2\}$ and $j \in \{i + 2, \dots, \text{diam}(D)\}$, $S_j \rightarrow_{D^c} S_i$.
4. $S_2 \leftrightarrow_{D^c} v$.
5. $S_5 \leftrightarrow_{D^c} v$.
6. $N^+(v) \subseteq S_3 \cup S_4$ and $v \rightarrow_D S_4$.

Proof. Let D , v and S_i as in the hypotheses. The first three properties follow from the definition of S_i .

4. From 2 we have that $S_2 \rightarrow_{D^c} v$. To prove that $v \rightarrow_{D^c} S_2$, we proceed by contradiction. Suppose that there is $x_2 \in S_2$ such that $v \rightarrow_D x_2$. By definition of S_2 , there is $x_1 \in S_1$ such that (x_2, x_1, v) is a path in D . It follows that (v, x_2, x_1, v) is an induced $\vec{C}_3$ in D , which is a contradiction. Therefore, $v \rightarrow_{D^c} S_2$.

5. From 2 we have that $S_5 \rightarrow_{D^c} v$. To prove that $v \rightarrow_{D^c} S_5$, we proceed by contradiction. Suppose that there is $x_5 \in S_5$ such that $v \rightarrow_D x_5$. By definition of S_5 , there is $x_i \in S_i$, with $i \in \{1, 2, 3, 4\}$, such that $(x_5, x_4, x_3, x_2, x_1, v)$ is a path in D . It follows that (v, x_5, x_4, x_3, x_2) is a 4-path in D . Since D is 4-quasi-transitive, there is an arc between v and x_2 , contradicting 4. Therefore, $v \rightarrow_{D^c} S_5$.
6. Since D is asymmetrical of diameter 4 or 5, and from parts 1, 4 and 5, we have that $N^+(v) \subseteq S_3 \cup S_4$. On the other hand, $v \rightarrow_D S_4$ follows from the definition of S_4 , part 2 and the asymmetry of D . $\square$

Lemma 23. *Let D be an asymmetrical 4-quasi-transitive digraph of diameter 4 or 5. Let v be a vertex of D and S_i be the set $\{x \in V(D) : d(x, v) = i\}$ with $i \in \{1, \dots, \text{diam}(D)\}$. If D is $\overrightarrow{C}_3$ -free and $\overrightarrow{C}_5$ -free, then the following properties hold.*

1. S_4 is an independent set.
2. $S_3 \cap N^+(v)$, induces a transitive tournament.
3. $S_3 \setminus N^+(v)$ is an independent set.
4. $(S_3 \cap N^+(v)) \rightarrow_D (S_3 \setminus N^+(v))$.
5. For every $x_4 \in S_4$ and every $x_3 \in S_3$, there is an arc between x_4 and x_3 .
6. If $x_4 \in S_4$ and $x_3, x'_3 \in S_3$ such that $x_4 \rightarrow_D x_3$ and $x_3 \rightarrow_D x'_3$, then $x_4 \rightarrow_D x'_3$.
7. S_3 is an independent set. Moreover, if $S_3 \cap N^+(v) \neq \emptyset$, then $|S_3| = 1$ and $S_4 \rightarrow S_3$.

Proof. Let D , v and S_i be as in the hypotheses.

1. Proceeding by contradiction, suppose that S_4 is not an independent set. Thus, there are vertices x_4 and x'_4 in S_4 such that $x_4 \rightarrow_D x'_4$. By definition of S_4 , there exists $x_i \in S_i$, with $i \in \{1, 2, 3\}$, such that (x_4, x_3, x_2, x_1, v) is a path in D . Moreover, by part 6 of Lemma 22, we have $v \rightarrow_D x_4$. It follows that (v, x_4, x'_4, x_3, x_2) is a 4-path in D . Since D is 4-quasi-transitive, there is an arc between v and x_2 , which is a contradiction. Hence S_4 is independent.
2. First, we prove that there is an arc between any two vertices in $S_3 \cap N^+(v)$. Let $u, w \in S_3 \cap N^+(v)$. By definition of S_3 , there is $x_i \in S_i$, with $i \in \{1, 2\}$, such that (u, x_2, x_1, v) is a path in D . It follows that (u, x_2, x_1, v, w) is a 4-path in D , and hence, there is an arc between u and w . Therefore, $S_3 \cap N^+(v)$ induces a semicomplete digraph in D , even more, since D is asymmetrical and $\overrightarrow{C}_3$ -free, we conclude that $S_3 \cap N^+(v)$ induces a transitive tournament.
3. Proceeding by contradiction suppose that there are vertices u and w in $S_3 \setminus N^+(v)$ such that $w \rightarrow_D u$. By definition of S_3 , there is $x_i \in S_i$, with $i \in \{1, 2\}$, such that (u, x_2, x_1, v) is a path in D . It follows that (w, u, x_2, x_1, v) is a 4-path in D . Since D is 4-quasi-transitive, there is an arc between w and v , contradicting part 1 of Lemma 22 or the choice of w . Therefore $S_3 \setminus N^+(v)$ is independent.
4. Let $u \in S_3 \setminus N^+(v)$ and $w \in S_3 \cap N^+(v)$. First we prove that there is an arc between u and w . By definition of S_3 , there is $x_i \in S_i$, with $i \in \{1, 2\}$, such that (u, x_2, x_1, v) is a path in D . It follows that (u, x_2, x_1, v, w) is a 4-path in D . Thus, there is an arc between u and w in D . If $u \rightarrow_D w$, then (u, w, x'_2, x'_1, v) is a 4-path in D , for some $x'_i \in S_i$ with $i \in \{1, 2\}$. Since D is a 4-quasi-transitive digraph, we have the existence of an arc between u and v which is impossible. Therefore, $w \rightarrow_D u$.
5. Let $x_4 \in S_4$ and $x_3 \in S_3$. On the one hand, by definition of S_3 , there is $x_i \in S_i$, with $i \in \{1, 2\}$, such that (x_3, x_2, x_1, v) is a path in D . On the other hand, by part 6 of Lemma 22, $v \rightarrow_D x_4$. It implies that (x_3, x_2, x_1, v, x_4) is a 4-path in D , thus there is an arc between x_4 and x_3 .
6. Let $x_4 \in S_4$ and $x_3, x'_3 \in S_3$ be such that $x_4 \rightarrow_D x_3$ and $x_3 \rightarrow_D x'_3$. By part 5 of this lemma, there is an arc between x'_3 and x_4 . If $x'_3 \rightarrow_D x_4$, then (x_4, x_3, x'_3, x_4) is an induced $\overrightarrow{C}_3$ in D , which is impossible. Therefore $x_4 \rightarrow_D x'_3$.
7. Observe that if $S_3 \cap N^+(v) = \emptyset$, then the result follows from part 3 of this Lemma. Suppose that $S_3 \cap N^+(v) \neq \emptyset$. Let $\{v_1, \dots, v_n\}$ be the acyclic ordering of the vertices of $S_3 \cap N^+(v)$. By part 5, there is an arc between v_1 and x_4 , for every vertex x_4 in S_4 . If there is $z \in S_4$ such that $v_1 \rightarrow_D z$, then, by definition of S_4 , there is $x_i \in S_i$, with $i \in \{1, 2, 3\}$, such that (x_4, x_3, x_2, x_1, v) is a path in D . Moreover, since D is asymmetrical, we have that (v, v_1, z, x_3, x_2) is a 4-path in D . It follows that there is an arc between v and x_2 , which is impossible by part 4 of Lemma 22. Hence, $S_4 \rightarrow_D v_1$. We prove that $S_3 = \{v_1\}$. Proceeding by contradiction, suppose that there is $u \in S_3 \setminus \{v_1\}$. By parts 2 and 4, we have that $u \in N^+(v_1)$, and by definition of S_3 , there is $x_2 \in S_2$ such that $u \rightarrow_D x_2$. It follows that (v, x_4, v_1, u, x_2) is a 4-path in D , thus there is an arc between v and x_2 , contradicting part 4 of Lemma 22. $\square$

Lemma 24. *The unique asymmetrical 4-quasi-transitive critical kernel imperfect digraph of diameter 4, is $\overrightarrow{C}_5$. Moreover, there are no asymmetrical 4-quasi-transitive critical kernel imperfect digraphs of diameter 5.*

Proof. Proceeding by contradiction, suppose that there is a 4-quasi-transitive asymmetrical critical kernel imperfect digraph D of diameter 4 or 5, different from $\overrightarrow{C}_5$. We know that D is strong, as well as $\overrightarrow{C}_3$ -free and $\overrightarrow{C}_5$ -free. Let v be a vertex of D such that the set $S_i = \{x \in V(D) : d(x, v) = i\}$ is not empty for each $i \in \{1, \dots, \text{diam}(D)\}$. Note that the hypotheses of Lemmas 22 and 23 hold. We consider two cases.

Case 1. Suppose first that $S_3 \cap N^+(v) \neq \emptyset$. By part 7 of Lemma 23, $|S_3| = 1$, so suppose that $S_3 = \{w\}$. Note that $v \rightarrow_D w$. First, we prove that $w \rightarrow S_2$. Let $x_2 \in S_2$. By definition of S_2 there is $x_1 \in S_1$, such that (x_2, x_1, v) is a path in D . By part 6

of [Lemmas 22](#) and part 7 of [Lemma 23](#), we have that $v \rightarrow_D S_4$ and $S_4 \rightarrow w$, and therefore it follows that (x_2, x_1, v, x_4, w) is a 4-path in D , with $x_4 \in S_4$. Thus there is an arc between x_2 and w . Note that, if $x_2 \rightarrow_D w$, then (x_2, w, x'_2, x'_1, v) is a 4-path in D , for some $x'_i \in S_i$ with $i \in \{1, 2\}$. Hence, there is an arc between x_2 and v , which is impossible by part 4 of [Lemma 22](#). Therefore, $w \rightarrow S_2$.

Now, we prove that, for every $x_1 \in S_1$ and every $x_2 \in S_2$, there is an arc between x_1 and x_2 . Let $x_1 \in S_1$ and $x_2 \in S_2$. Since $w \rightarrow_D S_2$, using part 6 of [Lemma 22](#) we have that for some $x_4 \in S_4$, the 4-path (x_1, v, x_4, w, x_2) exists in D and hence, there is an arc between x_1 and x_2 . Furthermore, we show that for every $x_1 \in S_1$, either $x_1 \rightarrow_D S_2$ or $S_2 \rightarrow_D x_1$. Suppose that this is not true, so there are $x_1 \in S_1$ and $x_2, x'_2 \in S_2$ such that $x_1 \rightarrow_D x_2$ and $x'_2 \rightarrow_D x_1$. Since D is asymmetrical, by definition of S_2 it follows that $(x'_2, x_1, x_2, x'_1, v)$ is a 4-path in D , for some $x'_1 \in S_1$. Hence, there is an arc between x_2 and v , contradicting part 4 of [Lemma 22](#).

Consider the sets $B = \{x \in S_1 : x \rightarrow_D S_2\}$ and $C = \{x \in S_1 : S_2 \rightarrow_D x\}$. Notice that $C \neq \emptyset$, $B \rightarrow_D S_2$ and $S_2 \rightarrow_D C$. We now show that, $B \rightarrow C$. Let $b \in B$ and $c \in C$. By definition of B and C , we have that (b, v, w, x_2, c) is a 4-path from b to c , for some $x_2 \in S_2$. This implies that there is an arc between b and c . If $c \rightarrow_D b$ then, for some $x_2 \in S_2$, we have that (b, x_2, c, b) is an induced $\vec{C}_3$ in D , which is impossible. Hence, $b \rightarrow_D c$. Now, we prove that $w \leftrightarrow_{D^c} C$. By part 3 of [Lemma 22](#) we have that $w \rightarrow_{D^c} C$. To prove $C \rightarrow_{D^c} w$, aiming for a contradiction, we suppose that there is $c \in C$ such that $c \rightarrow_D w$. It follows that there exists $x_2 \in S_2$ such that (c, w, x_2, c) is an induced $\vec{C}_3$ in D , contradicting the fact that D is $\vec{C}_3$ -free. Therefore, $w \leftrightarrow_{D^c} C$. We affirm that $|C| = 1$. Indeed, suppose for the sake of contradiction, that there are $c_1, c_2 \in C$. It follows that (c_1, v, w, x_2, c_2) is a 4-path in D , hence there is an arc between c_1 and c_2 . Suppose without loss of generality that $c_1 \rightarrow c_2$, then (c_1, c_2, v, x_4, w) is a 4-path in D , for some $x_4 \in S_4$. But this implies that there is an arc between c_1 and w , contradicting the fact that $w \leftrightarrow_{D^c} C$. Thus $|C| = 1$.

Case 1.1 If D has diameter 4, then we claim that $N = \{w, c\}$ is a kernel of D , with c the unique element of C . We have already proved that $w \leftrightarrow_{D^c} c$, and hence, N is independent. To prove that N is absorbent, it suffices to observe that $S_4 \rightarrow_D w$, $v \rightarrow_D w$, $S_2 \rightarrow_D c$, and $B \rightarrow_D c$. Therefore, N is a kernel of D which is a contradiction.

Case 1.2 Suppose that D has diameter 5, and let $D_5 = D[S_5]$. Since D is critical kernel imperfect, D_5 has a kernel, say N_5 . We claim that $N = \{w, c\} \cup N_5$ is a kernel of D . To prove the independence, note that, by definition, N_5 is an independent set of D_5 and, in consequence, of D . We have that $w \rightarrow_{D^c} c$, and, by part 3 of [Lemma 22](#), $N_5 \rightarrow_{D^c} w$ and $N_5 \rightarrow_{D^c} c$. Thus we have to prove that $w \rightarrow_{D^c} N_5$ and $c \rightarrow_{D^c} N_5$. Let $x_5 \in N_5$. If $w \rightarrow_D x_5$, then, as a consequence of part 7 of [Lemma 23](#), for some $x_4 \in S_4$ we have that (w, x_5, x_4, w) is an induced $\vec{C}_3$ in D , which is impossible. Hence $w \rightarrow_{D^c} x_5$. On the other hand, if $c \rightarrow_D x_5$, then (v, w, x_2, c, x_5) is a 4-path in D , for some $x_2 \in S_2$, which is not possible. Therefore, $c \rightarrow_{D^c} x_5$, and N is independent. To prove that N is absorbent, it is sufficient to note that $S_4 \rightarrow_D w$, $v \rightarrow_D w$, $S_2 \rightarrow_D c$, $B \rightarrow_D c$, and N_5 is absorbent in D_5 . Therefore, N is a kernel of D , contradicting the fact that D is critical kernel imperfect.

Case 2. Now, suppose that $S_3 \cap N^+(v) = \emptyset$. By part 7 of [Lemma 23](#), S_3 is an independent set of D . Since $S_3 \cap N^+(v) = \emptyset$, part 6 of [Lemma 22](#) implies $v \leftrightarrow_{D^c} S_3$ and $N^+(v) = S_4$. First, observe that if there are $x_2 \in S_2$ and $x_3 \in S_3$ such that $x_2 \rightarrow_D x_3$, then (x_2, x_3, x'_2, x_1, v) is a 4-path in D , for some $x'_2 \in S_2$. Therefore, there is an arc between x_2 and v , which is impossible by part 4 of [Lemma 22](#). Therefore, $S_2 \rightarrow_{D^c} S_3$.

Now, we prove that for every $x_3 \in S_3$ we have that either $x_3 \rightarrow_D S_4$ or $S_4 \rightarrow_D x_3$. Let $x_3 \in S_3$ and, in order to reach a contradiction, suppose that there are $x_4, x'_4 \in S_4$ such that $x_3 \rightarrow_D x_4$ and $x'_4 \rightarrow_D x_3$. It follows that $(v, x'_4, x_3, x_4, x'_3)$ is a 4-path in D , for some $x'_3 \in S_3$, and thus there is an arc between v and x'_3 , contradicting the fact that $v \leftrightarrow_{D^c} S_3$. Hence, for every $x_3 \in S_3$ we have that either $x_3 \rightarrow_D S_4$, or $S_4 \rightarrow_D x_3$. Consider the sets $U = \{x \in S_3 : x \rightarrow_D S_4\}$ and $W = \{x \in S_3 : S_4 \rightarrow_D x\}$. Notice that $W \neq \emptyset$, $U \rightarrow_D S_4$, and $S_4 \rightarrow_D W$.

Let $D_2 = D[S_2]$. Since D is a critical kernel imperfect digraph, D_2 has a kernel, say N_2 . We claim that for every $x_3 \in S_3$ there is $n \in N_2$ such that $x_3 \rightarrow_D n$, and moreover, $W \rightarrow_D N_2$. Let $x_3 \in S_3$, it follows that there is $x_2 \in S_2$ such that $x_3 \rightarrow_D x_2$. If $x_2 \notin N_2$, then there is $n \in N_2$ such that $x_2 \rightarrow_D n$. It follows that (x_3, x_2, n, x_1, v) is a 4-path, for some $x_1 \in S_1$, and hence, there is an arc between x_3 and v , which is impossible. Thus, $x_2 \in N_2$. To prove the second claim, assume that $x_3 \in W$. Let $z \in N_2$, it follows that (z, x_1, v, x_4, x_3) is a 4-path in D , for some $x_1 \in S_1$ and with $x_4 \in S_4$, thus there is an arc between z and x_3 . Since $S_2 \rightarrow_{D^c} S_3$, we have $x_3 \rightarrow_D z$. Therefore, $W \rightarrow_D N_2$.

Finally, we claim that $N = S_4 \cup N_2$ is a kernel of D . To prove that N is independent, observe that, on the one hand, N_2 is already an independent set, and on the other hand, part 1 of [Lemma 23](#) implies that S_4 is independent. Also, an application of part 3 of [Lemma 22](#) implies $S_4 \rightarrow_{D^c} N_2$. So, it remains to prove that $N_2 \rightarrow_{D^c} S_4$. For the sake of contradiction, suppose that there are $x_4 \in S_4$ and $x_2 \in N_2$ such that $x_2 \rightarrow_D x_4$. This implies that (x_2, x_4, x_3, x_2) is an induced $\vec{C}_3$ in D , for some $x_3 \in W$, contradicting that D is $\vec{C}_3$ -free. Hence $N_2 \rightarrow_{D^c} S_4$. We conclude that N is independent.

For the absorbance of N , we have that $v \rightarrow_D S_4$, also N_2 is a kernel of D_2 , and for every $x_3 \in S_3$ there is $n \in N_2$ such that $x_3 \rightarrow_D n$. When D has diameter 5, we also have that, for every $x_5 \in S_5$, there is $x_4 \in S_4$ such that $x_5 \rightarrow_D x_4$. So, it remains to prove that for every $x_1 \in S_1$ there is $x \in N$ such that $x_1 \rightarrow_D x$. Let $x_1 \in S_1$ and $x \in N_2$. Since (x_1, v, x_4, w, x) is a 4-path in D , for some $x_4 \in S_4$, then there is an arc between x_1 and x . If $x_1 \rightarrow_D x$, then we have the desired result. Suppose that $x \rightarrow_D x_1$, it follows that $\gamma = (x, x_1, v, x_4, w, x)$ is a 5-cycle of D , with $x_4 \in S_4$ and $w \in W$. Since D is $\vec{C}_5$ -free, γ has at least one diagonal. Observe that $x_4 \leftrightarrow_{D^c} x$, $x \leftrightarrow_{D^c} v$, $v \leftrightarrow_{D^c} w$ and $x_4 \rightarrow_{D^c} x_1$, it follows that $x_1 \rightarrow_D x_4$, with $x_4 \in S_4$. Hence N is an absorbent set in D . Thus N is a kernel of D , which is a contradiction.

From above, we conclude that the unique asymmetrical 4-quasi-transitive critical kernel imperfect digraph of diameter 4 is $\vec{C}_5$, and there are no asymmetrical 4-quasi-transitive critical kernel imperfect digraphs of diameter 5. $\square$

We are ready to prove the main result of this section.

Proof of Theorem 10. The result follows from Corollary 17 and Lemmas 18, 21 and 24. $\square$

As a consequence of Theorem 10 we have the following result.

Corollary 25. *If D is an asymmetrical 4-quasi-transitive digraph, then it is kernel-perfect if and only if it is $\{\vec{A}_3, \vec{C}_5\}$ -free.*

4. 4-transitive digraphs

In [21], Hernández-Cruz gave a structural characterization of strong 4-transitive digraphs. Note that 4 is the largest value of k such that strong k -transitive digraphs are characterized.

Theorem 26 ([21]). *Let D be a strong 4-transitive digraph. Then exactly one of the following possibilities holds*

1. D is a complete digraph.
2. D is a 3-cycle extension.
3. D has circumference 3, a 3-cycle extension as a spanning subdigraph with cyclical partition $\{V_0, V_1, V_2\}$, at least one symmetrical arc exists in D and for every symmetrical arc $(v_i, v_{i+1}) \in A(D)$, with $v_j \in V_j$ for $j \in \{i, i+1\} \pmod{3}$, $|V_i| = 1$ or $|V_{i+1}| = 1$.
4. D has circumference 3, $UG(D)$ is not 2-edge-connected and $S_1, S_2, \dots, S_r$ are the vertex sets of the maximal 2-edge connected subgraphs of $UG(D)$, with $S_i = \{u_i\}$ for every $2 \leq i \leq r$ and such that $D[S_1]$ has a 3-cycle extension with cyclical partition $\{V_0, V_1, V_2\}$ as a spanning subdigraph. A vertex $v_0 \in V_0$ (without loss of generality) exists such that $v_0 \leftrightarrow_D u_j$, for every $2 \leq j \leq r$. Also $|V_0| = 1$ and $D[S_1]$ has the structure described in 1. or 2., depending on the existence of symmetrical arcs.
5. A complete biorientation of a 5-cycle.
6. D is a complete biorientation of the star $K_{1,r}$, $r \geq 3$.
7. D is a complete biorientation of a tree of diameter 3.
8. D is a strong digraph of order less than or equal to 4 not included in the previous families.

With the characterization in Theorem 26 we determine the 4-transitive critical kernel imperfect digraphs. As stated in Theorem 11, a 4-transitive digraph D is critical kernel imperfect if $D \cong \vec{A}_3$ or $D \cong \vec{A}_4$.

Proof of Theorem 11. Let D be a 4-transitive critical kernel imperfect digraph. Clearly, D satisfies exactly one of the possibilities of Theorem 26. It is not difficult to see that if a digraph is in the family 1, 5, 6, or 7 of Theorem 26, then the digraph has a kernel. Moreover, it is a kernel-perfect digraph. Hence, D is in family 2, 3, 4, or 8. Notice that $\vec{C}_3$ is in family 2, and every digraph which is in family 2, 3, or 4, contains $\vec{C}_3$ as an induced subdigraph. Hence, we have that if D is in family 2, then $D \cong \vec{C}_3$, and D is not in family 3 or 4.

Suppose that D is in family 8. By Theorem 1, if D is an asymmetrical digraph, then $D \cong \vec{C}_3$. Even more, by Theorem 2, if D is a semicomplete digraph, then $D \cong \vec{C}_3$ or $D \cong \vec{A}_4$. Now assume that D is not an asymmetrical and D is not semicomplete. As every digraph with at most 3 vertices is kernel-perfect or is $\vec{C}_3$, it follows that D has 4 vertices. Let $\{u, v, z, w\}$ be the vertex set of D . Since it is not semicomplete, the independence number of D is at least 2, but it must be exactly 2, as otherwise D has a kernel. Suppose without loss of generality that there is no arc between u and v . Since D has no kernel, $\{u, v\}$ is not an absorbent set. We can assume that (w, u) and (w, v) are not arcs of D . Moreover, since $\{u, v\}$ is independent and D is strong, it follows that $(w, z), (z, u), (z, v) \in A(D)$. We know that $\{z\}$ is not a kernel of D , and thus, (u, z) or (v, z) is not an arc of D . Suppose, without loss of generality, that $(v, z) \notin A(D)$, it follows that $(v, w) \in A(D)$. Since D is critical kernel imperfect, it has no $\vec{C}_3$ as induced subdigraph. Thus $(z, w) \in A(D)$. Note that $(u, w) \notin A(D)$, otherwise $\{w\}$ is a kernel of D . It follows that $\{u, w\}$ is a kernel of D , contradicting the fact that D is a critical kernel imperfect digraph. Hence D has to be a semicomplete or an asymmetrical digraph. $\square$

As a consequence of Theorem 11, we have the following result.

Theorem 27. *If D is a 4-transitive digraph, then D is kernel-perfect if and only if it is $\{\vec{A}_3, \vec{A}_4\}$ -free.*

5. Antitransitive digraphs

Let k be an integer, $k \geq 2$. A digraph D is k -anti-transitive if for every pair of distinct vertices u and v of D , the existence of a k -path from u to v in D implies that (u, v) is not an arc of D . We denote by TT_3 the asymmetrical transitive tournament with order 3.

Lemma 28. Let D be a 2-anti-transitive digraph, and let v be a vertex of D . The following statements hold

1. D has no TT_3 as subdigraph.
2. $N^+(v)$ and $N^-(v)$ are independent sets in D .

Proof. Let D be a 2-anti-transitive digraph. Note that if TT_3 is a subdigraph of D , then there are vertices x and y such that there is a 2-path from x to y and (x, y) is an arc of D , which is impossible by the definition of 2-anti-transitive digraph. For the second item, if v is a vertex of D and there is an arc between two vertices of $N^+(v)$ ($N^-(v)$), then TT_3 is a subdigraph of D , which is also impossible by 1. $\square$

Lemma 28 is the only tool we require to prove Theorem 12, which states that the only 2-anti-transitive critical kernel imperfect digraph of diameter 2 is $\overrightarrow{A}_3$.

Proof of Theorem 12. Let D be a 2-anti-transitive critical kernel imperfect digraph of diameter 2. If D is isomorphic to $\overrightarrow{A}_3$, there is nothing to prove. Proceeding by contradiction, suppose that $D \not\cong \overrightarrow{A}_3$. Being critical kernel imperfect, D is $\overrightarrow{A}_3$ -free.

Let v be a vertex of D . Consider the sets $S_i = \{x \in V(D) : d(x, v) = i\}$ with $i \in \{1, 2\}$. By hypothesis $V(D) = \{v\} \cup S_1 \cup S_2$. Moreover, since D has no kernel, $S_2 \neq \emptyset$. By Lemma 28 it follows that $S_1 = N^-(v)$ is an independent set of D . We claim that $N^+(v) \subseteq S_1$. Let x be a vertex in $N^+(v)$. If $x \notin S_1$, then $x \in S_2$. Thus there is $w \in S_1$ such that (x, w, v) is a 2-path in D , which implies that $\gamma = (x, w, v, x)$ is a 3-cycle in D . Since D is $\overrightarrow{A}_3$ -free, there is a symmetrical arc in γ , it follows that TT_3 is a subdigraph of D , which is impossible. Hence, $N^+(v) \subseteq S_1$.

By Lemma 28, $S_1 = N^-(v)$ is an independent set of D . Let $u \in V(D) \setminus S_1$. If $u \in S_2$, then there is $w \in S_1$ such that $u \rightarrow_D w$. On the other hand, D is strong, and since $N^+(v) \subseteq S_1$, if $u = v$, then is $w \in S_1$ such that $x \rightarrow_D w$. Therefore, S_1 is an absorbent set and a kernel of D , which is a contradiction. $\square$

Lemma 29. Let D be an asymmetrical $\overrightarrow{C}_3$ -free, 4-anti-transitive digraph of diameter 3. If $v \in V(D)$, then the following statements hold.

1. $d_D(w, v) = 3$, for each $w \in N^+(v)$.
2. $N^+(v)$ is an independent set.
3. $N^-(v)$ is an independent set.

Proof. Let w be in $N^+(v)$. Since D is an asymmetrical digraph of diameter 3, we have $1 < d_D(w, v) \leq 3$, moreover if $d_D(w, v) = 2$, then $\overrightarrow{C}_3$ is an induced subdigraph of D , which is impossible. Hence $d_D(w, v) = 3$.

For the second statement, let w and u be two vertices of $N^+(v)$. By the first statement, we have that $d_D(w, v) = 3$. Thus there is (w, z, x, v) a path from w to v in D . It implies that (w, z, x, v, u) is a path from w to u in D of length 4. Since D is a 4-anti-transitive digraph, (w, u) is not an arc of D . Therefore, $N^+(v)$ is an independent set.

For the third statement, let x and y be two vertices of $N^-(v)$. Note that $v \in N^+(x)$, by the first statement it follows that $d_D(v, x) = 3$. Thus there is (v, z, u, x) a path from v to x in D . It follows that (y, v, z, u, x) is a path from y to x in D of length 4. Since D is a 4-anti-transitive digraph, (y, x) is not an arc of D . Therefore, $N^-(v)$ is an independent set. $\square$

Lemma 29 enables us to prove Theorem 13, which states that the unique asymmetrical 4-anti-transitive critical kernel imperfect digraph of diameter 3 is $\overrightarrow{C}_5$.

Proof of Theorem 13. Let D be an asymmetrical 4-anti-transitive critical kernel imperfect digraph of diameter 3. Looking for a contradiction, suppose that $D \not\cong \overrightarrow{C}_5$. Since it is critical kernel imperfect, D is $\overrightarrow{C}_{2n+1}$ -free, for every $n \in \mathbb{N}$.

Let v_0 be a vertex of D such that the sets $S_i = \{x \in V(D) : d(x, v_0) = i\}$ are nonempty for $i \in \{1, 2, 3\}$. By hypothesis $V(D) = \{v_0\} \cup S_1 \cup S_2 \cup S_3$. Note that $S_1 = N^-(v_0)$ and, by Lemma 29, S_1 is an independent set of D . By definition $S_i \rightarrow_{D^c} v_0$ with $i \in \{2, 3\}$. Moreover, there is no arc from v_0 to y , for every $y \in S_2$, otherwise $\overrightarrow{C}_3$ is an induced subdigraph of D , which is impossible. From above, it follows that $N^+(v_0) \subseteq S_3$, moreover, since D is strong, then $N^+(v_0)$ is nonempty.

We claim that $S_1 \leftrightarrow_{D^c} S_3$. By definition, $S_3 \rightarrow_{D^c} S_1$. To prove $S_1 \rightarrow_{D^c} S_3$, we proceed by contradiction. Suppose that there are $x \in S_1$ and $y \in S_3$ such that (x, y) is an arc of D . Since $y \in S_3$, there are $u \in S_2$ and $w \in S_1$ such that (y, u, w, v_0) is a path in D . Observe that $w \neq x$, otherwise (y, u, x, y) is an induced $\overrightarrow{C}_3$ in D , which is impossible. It follows that (x, y, u, w, v_0) is a path in D of length 4. Since D is 4-anti-transitive, we have that (x, v_0) is not an arc of D , which is a contradiction. Therefore, $S_1 \leftrightarrow_{D^c} S_3$.

It follows from Lemma 29 that $N^+(v_0) \subseteq S_3$. We now prove that $(S_3 \setminus N^+(v_0)) \leftrightarrow_{D^c} N^+(v_0)$. Let $x \in N^+(v_0)$ and $y \in S_3 \setminus N^+(v_0)$. By definition, there are $u \in S_2$ and $w \in S_1$ such that (y, u, w, v_0) is a path in D and, even more (y, u, w, v_0, x) is a path in D of length 4. Since D is 4-anti-transitive, we have that (y, x) is not an arc in D . To prove that (x, y) is not an arc of D suppose, for the sake of contradiction, that $(x, y) \in A(D)$. By definition of S_3 , there are $u' \in S_2$ and $w' \in S_1$ such that (y, u', w', v_0) is a path in D . It follows that $\gamma = (v_0, x, y, u', w', v_0)$ is a 5-cycle in D . We know that $S_1 \leftrightarrow_{D^c} S_3$, thus $x \leftrightarrow_{D^c} w'$ and $y \leftrightarrow_{D^c} w'$. Since $S_2 \leftrightarrow_{D^c} v_0$, the arc (u', v_0) is present in D^c . By definition of $S_3 \setminus N^+(v_0)$, we also have

$v_0 \leftrightarrow_{D^c} y$. In addition, (u', x) is not an arc of D , because D is $\vec{C}_3$ -free. Even more, (x, u') is neither an arc of D , because we would have $y, u' \in N^+(x)$, and Lemma 29 states that $N^+(x)$ is an independent set, but (y, u') is an arc of D . Thus, $x \leftrightarrow_{D^c} u'$, and we conclude that γ is an induced $\vec{C}_5$ in D , which is a contradiction. Therefore $(S_3 \setminus N^+(v_0)) \leftrightarrow_{D^c} N^+(v_0)$.

Let D_3 the subdigraph induced by $S_3 \setminus N^+(v_0)$. Since D is critical kernel imperfect, D_3 has a kernel, say K_3 . We claim that $K = S_1 \cup N^+(v_0) \cup K_3$ is a kernel of D . Note that $N^+(v_0) \cup K_3 \subseteq S_3$ and $S_3 \leftrightarrow_{D^c} S_1$, moreover $K_3 \subseteq (S_3 \setminus N^+(v_0))$ and $(S_3 \setminus N^+(v_0)) \leftrightarrow_{D^c} N^+(v_0)$. Since $S_1 = N^-(v_0)$, by the definition of K_3 , and by Lemma 29, we obtain that K is an independent set of D . To prove that K is absorbent, let $u \in V(D) \setminus K$. If $u = v_0$, then $v_0 \rightarrow N^+(x_0)$. If $u \in S_2$, then, by definition there is $w \in S_1$, such that (u, w) is an arc of D . And, if $u \in S_3$, then $u \in ((S_3 \setminus N^+(v_0)) \setminus N_3)$ and, by the definition of K_3 , there is $x \in K_3$ such that (u, x) is an arc of D . Therefore, K is a kernel of D , which is a contradiction. We conclude that $\vec{C}_5$ is the unique 4-anti-transitive critical kernel imperfect digraph of diameter 3. $\square$

6. Conclusions

We characterized all the critical kernel imperfect digraphs which are 4-quasi-transitive oriented graphs. There are only two of them: $\vec{A}_3$ and $\vec{C}_5$. As we mentioned in the introduction, every semicomplete digraph is k -quasi-transitive for every positive integer k greater than 1. Hence, $\vec{A}_n$ is a critical kernel imperfect 4-quasi-transitive digraph for every n at least 3. So, there is an infinite family of critical kernel imperfect 4-quasi-transitive digraphs when we admit symmetric arcs. It is natural to ask, are the directed antihole the only such critical kernel imperfect digraphs?

Problem 30. Characterize all the 4-quasi-transitive critical kernel imperfect digraphs. In particular, is it true that such digraphs are either $\vec{C}_5$ or a directed antihole?

In a similar direction, we were able to characterize all the critical kernel imperfect 4-transitive digraphs. It turns out that the complete list only contains $\vec{A}_3$ and $\vec{A}_4$. Again, it is clear that $\vec{A}_5$ is a 5-transitive digraph which is critical kernel imperfect. Could it be the case that all k -transitive critical kernel imperfect digraphs are directed antihole?

Question 31. Is it true that every k -transitive critical kernel imperfect digraph is a directed antihole?

The lack of knowledge of the structure of k -anti-transitive digraphs prevented us from getting more general results for the families of 2- and 4-anti-transitive digraphs. We propose the problem of finding nice structural descriptions of the families of k -anti-transitive digraphs, at least for some small values of k . Naturally, it would also be nice to know the complete families of critical kernel imperfect 2-anti-transitive digraphs. Notice that $\vec{C}_7(\{1, 2\})$ is a 3-anti-transitive asymmetric digraph having diameter 3, so it is no longer the case that the only critical kernel imperfect digraphs are directed odd cycles or directed antihole.

Data availability

No data was used for the research described in the article.

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